Week 5 complete solutions

Problem-sheet solutions

Each solution identifies its outcome, intended method, conditions, derivation, conclusion, mark allocation and common mistakes.

`W05-Q01-PA (M7)`

Method and conditions. Differentiate on the stated domain $p > 0, w > 0$. The gradient is a 2×1 column, and the total differential is its transpose multiplied by the change column.

$$\nabla c(p, w) = \begin{bmatrix} 2p/w \\ -p^2/w^2 + 2/w \end{bmatrix}.$$

Thus

$$\mathrm{d}c = \frac{2p}{w}\,\mathrm{d}p + \left(-\frac{p^2}{w^2} + \frac{2}{w}\right)\mathrm{d}w.$$

At $(p, w) = (2, 1)$,

$$\nabla c(2, 1) = \begin{bmatrix} 4 \\ -2 \end{bmatrix}.$$

Therefore,

$$\Delta c \approx \mathrm{d}c = 4(0.02) - 2(-0.01) = 0.10.$$

Final result. The total differential is the expression above, and the first-order approximate change is 0.10.

Marks (5). Correct c_p, 1; correct c_w, 1; correctly oriented total differential, 1; gradient at the base point, 1; approximation, 1.

Common mistakes. Writing the gradient as a row; differentiating p^2/w incorrectly with respect to w; treating the differential as an exact finite change.

`W05-Q01-PB (M7)`

Method and conditions. Define $G(x, y, \theta) = x^2 y + \theta \log y$. The composition is differentiable near $\theta = 0$ because $y(0) = 2 > 0$, and $y(\theta) = 2 - \theta$ remains positive for θ sufficiently close to zero. Use all three channels of change:

$$\frac{\mathrm{d}z}{\mathrm{d}\theta} = G_x x' + G_y y' + G_\theta.$$

The partial derivatives are

$$G_x = 2xy, \qquad G_y = x^2 + \frac{\theta}{y}, \qquad G_\theta = \log y.$$

Also, $x' = 1$ and $y' = -1$. At $\theta = 0$, $x = 1$ and $y = 2$, so

$$\left.\frac{\mathrm{d}z}{\mathrm{d}\theta}\right|_{\theta=0} = 4(1) + 1(-1) + \log 2 = 3 + \log 2.$$

Final result.

$$\left.\frac{\mathrm{d}z}{\mathrm{d}\theta}\right|_{\theta=0} = 3 + \log 2,$$

valid locally where $y(\theta) > 0$.

Marks (6). Domain condition, 1; complete total-derivative formula, 2; three correct partial derivatives, 1; correct path derivatives and point values, 1; final derivative, 1.

Common mistakes. Omitting the direct G_θ term; differentiating $\theta \log y(\theta)$ as only $\log y$; ignoring the positive-logarithm domain.

`W05-Q02-PA (M7)`

Method and conditions. The polynomial F is continuously differentiable. The scalar implicit-function condition is $F_x \neq 0$ at the point.

$$F_x(x, \theta) = 2x + \theta, \qquad F_\theta(x, \theta) = x.$$

At $(2, 1)$,

$$F_x(2, 1) = 5 \neq 0.$$

Hence $x(\theta)$ exists locally and

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -\frac{F_\theta}{F_x}.$$

Evaluating gives

$$\left.\frac{\mathrm{d}x}{\mathrm{d}\theta}\right|_{(2,1)} = -\frac{2}{5}.$$

Final result. The required local condition holds, and the implicit derivative is $-2/5$.

Marks (5). Differentiability and $F_x \neq 0$ condition, 1; correct partials, 1; verified non-zero value, 1; implicit formula, 1; derivative, 1.

Common mistakes. Checking $F_\theta \neq 0$ instead of $F_x \neq 0$; omitting the minus sign; evaluating the partial derivatives at the wrong point.

`W05-Q02-PB (M7)`

Method and conditions. Let $\mathbf{F} = [F_1, F_2]^\mathsf{T}$. The Jacobian with respect to the 2×1 endogenous vector $\mathbf{x} = [x, y]^\mathsf{T}$ is 2×2:

$$\mathbf{J}_{\mathbf{F},\mathbf{x}} = \begin{bmatrix} 1 & 1 \\ 2x & 2y \end{bmatrix}.$$

At $(1, 2, 0)$,

$$\mathbf{J}_{\mathbf{F},\mathbf{x}} = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}, \qquad \det(\mathbf{J}_{\mathbf{F},\mathbf{x}}) = 2 \neq 0.$$

The functions are continuously differentiable and the Jacobian is nonsingular, so the system defines $\mathbf{x}(\theta)$ locally. Moreover,

$$\mathbf{F}_\theta = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Differentiating $\mathbf{F}(\mathbf{x}(\theta), \theta) = \mathbf{0}$ gives

$$\begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x'(0) \\ y'(0) \end{bmatrix} = - \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Thus

$$x'(0) + y'(0) = -1, \qquad 2x'(0) + 4y'(0) = 1.$$

Subtracting twice the first equation from the second yields $2y'(0) = 3$. Therefore,

$$x'(0) = -\frac{5}{2}, \qquad y'(0) = \frac{3}{2}.$$

Final result.

$$\left.\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}\theta}\right|_{\theta=0} = \begin{bmatrix} -5/2 \\ 3/2 \end{bmatrix}.$$

Marks (7). Correct 2×2 Jacobian and dimensions, 1; determinant and nonsingularity, 1; regularity/local-existence statement, 1; correct $\mathbf{F}_\theta$, 1; differentiated system, 1; solution of the system, 2.

Common mistakes. Inverting a 2×3 full Jacobian; omitting the nonsingularity check; failing to negate $\mathbf{F}_\theta$; reporting row rather than column derivatives.

`W05-Q03-PA (M7)`

Method and conditions. A function is homogeneous only if one common power of t factors from the whole function when every argument is scaled by the same $t > 0$. For each differentiable homogeneous function, verify Euler's identity on its stated positive domain.

$$f(tx, ty) = (tx)^{1/3}(ty)^{2/3} = t^{1/3+2/3} f(x, y) = t f(x, y).$$

Thus f is homogeneous of degree one. Its partial derivatives are

$$f_x = \frac{1}{3} x^{-2/3} y^{2/3}, \qquad f_y = \frac{2}{3} x^{1/3} y^{-1/3}.$$

Therefore,

$$x f_x + y f_y = \frac{1}{3} f(x, y) + \frac{2}{3} f(x, y) = f(x, y).$$

For r,

$$r(tx, ty) = tx + t^2 y^2.$$

The two terms scale with different powers. There is no common k for which this equals $t^k(x+y^2)$ for all x, y and $t>0$. Hence r is not homogeneous.

For h,

$$h(tx,ty)=\frac{t^2x^2}{t(x+y)}=th(x,y),$$

so h is homogeneous of degree one on the positive domain. Its partial derivatives are

$$h_x=\frac{x(x+2y)}{(x+y)^2},\qquad h_y=-\frac{x^2}{(x+y)^2}.$$

At $(2,1)$,

$$h_x=\frac{8}{9},\qquad h_y=-\frac{4}{9},\qquad h(2,1)=\frac{4}{3}.$$

Thus

$$xh_x+yh_y=2\left(\frac{8}{9}\right)-\frac{4}{9}=\frac{4}{3}=h(2,1),$$

as Euler's theorem requires for degree one.

Final result. The function f is homogeneous of degree one and satisfies $xf_x+yf_y=f$. The function r is not homogeneous. The function h is homogeneous of degree one and satisfies Euler's identity at $(2,1)$.

Marks (10). Scaling, degree and Euler verification for f, 4; valid disproof for r, 2; scaling and degree for h, 1; partial derivatives of h, 1; evaluated terms, 1; Euler verification for h, 1.

Common mistakes. Scaling only one argument of f; calling a sum homogeneous when its terms have different degrees; scaling the denominator of h by t^2; omitting the degree on the right-hand side of Euler's identity.

`W05-Q04-PA (M8)`

Method and conditions. Use the symmetric part of the given matrix because the skew-symmetric part contributes zero to a quadratic form.

$$\mathbf{S}=\frac{\mathbf{A}+\mathbf{A}^{\mathsf{T}}}{2}=\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}.$$

The leading principal minors are

$$2>0,\qquad \det(\mathbf{S})=4-1=3>0.$$

Therefore $\mathbf{S}$ is positive definite. Equivalently,

$$q(\mathbf{x})=2x^2+2xy+2y^2=(x+y)^2+x^2+y^2>0$$

for every non-zero $\mathbf{x}$.

Final result. The representing symmetric matrix is $\mathbf{S}$ above, and q is positive definite.

Marks (5). Correct symmetric part, 2; valid definiteness test, 1; correct calculations, 1; classification, 1.

Common mistakes. Applying a symmetric-matrix test directly to $\mathbf{A}$; averaging only the off-diagonal entries incorrectly; confusing positive determinant with positive definiteness without checking the first minor.

`W05-Q04-PB (M8)`

Method and conditions. Because $\mathbf{B}(a)$ is symmetric, classify the quadratic form from its eigenvalue signs.

The eigenvectors $[1,1]^\mathsf{T}$ and $[1,-1]^\mathsf{T}$ have eigenvalues

$$\lambda_1 = 2 + a, \qquad \lambda_2 = 2 - a,$$

respectively.

- If $-2 < a < 2$, both eigenvalues are positive, so the form is positive definite.
- If $a = 2$, the eigenvalues are 4 and 0. If $a = -2$, they are 0 and 4. In both boundary cases the form is positive semidefinite but not positive definite.
- If $|a| > 2$, one eigenvalue is positive and the other negative, so the form is indefinite.

The trace is $4 > 0$ for every a, so both eigenvalues cannot be non-positive. Negative definite and negative semidefinite cases therefore do not occur.

Final result.

Parameter range	Classification
$\lvert a\rvert < 2$	positive definite
$\lvert a\rvert = 2$	positive semidefinite, not positive definite
$\lvert a\rvert > 2$	indefinite

Marks (7). Two eigenvalues, 2; positive-definite range, 1; two boundary cases, 2; indefinite range, 1; exclusion of other classes, 1.

Common mistakes. Omitting $a = -2$; calling a zero-eigenvalue boundary positive definite; using only the determinant and failing to distinguish the boundary from the interior.

`W05-Q05-PA (M8)`

Method and conditions. Use the convex-combination definition for C and disprove convexity of D with two feasible points whose midpoint is infeasible.

Take $(x_1, y_1), (x_2, y_2) \in C$ and $\lambda \in [0,1]$. Then $y_i \geq x_i^2$, and

$$\begin{aligned}\big(\lambda x_1 + (1-\lambda)x_2\big)^2 &\leq \lambda x_1^2 + (1-\lambda)x_2^2 \\ &\leq \lambda y_1 + (1-\lambda)y_2.\end{aligned}$$

Thus every convex combination belongs to C, so C is convex.

For D, both $(-1,1)$ and $(1,1)$ satisfy $y \leq x^2$. Their midpoint is $(0,1)$, but $1 \leq 0^2$ is false. Therefore D is not convex.

Final result. C is convex; D is not convex.

Marks (6). Correct convex-combination setup, 1; square-function inequality, 1; feasibility conclusion for C, 1; two feasible points in D, 1; infeasible midpoint, 1; two classifications, 1.

Common mistakes. Treating the region below every convex graph as convex; checking only one line segment in C; giving points for D whose midpoint remains feasible.

`W05-Q05-PB (M8)`

Method and conditions. The positive orthant is convex. Use the Hessian for convexity and concavity, then use convex upper contour sets for quasiconcavity.

$$\mathbf{H}_g(x,y) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Its eigenvalues are 1 and -1, so it is indefinite. Hence g is neither convex nor concave on $\mathbb{R}^2_{++}$.

For $c > 0$, the upper contour set is

$$\{(x,y) \in \mathbb{R}^2_{++} : xy \geq c\} = \{(x,y) : y \geq c/x,\ x > 0\}.$$

Since

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2}\left(\frac{c}{x}\right) = \frac{2c}{x^3} > 0,$$

this is the epigraph of a convex function and is therefore convex. For $c \leq 0$, every point in the positive orthant satisfies $xy > 0 \geq c$, so the upper contour set is the whole convex domain. Every upper contour set is convex, and g is quasiconcave.

Final result. g is neither convex nor concave, but it is quasiconcave.

Marks (6). Correct Hessian, 1; indefinite classification, 1; neither convex nor concave conclusion, 1; upper contour representation, 1; convexity argument for $c > 0$, 1; $c \leq 0$ case and quasiconcavity conclusion, 1.

Common mistakes. Treating an indefinite Hessian as a failure of quasiconcavity; checking lower rather than upper contour sets; omitting non-positive contour levels.

`W05-Q06-PA (S5)`

Method and conditions. Sum the joint PMF over the other variable. The six entries are non-negative and sum to one.

The marginal PMF of X is

$$p_X(0) = \frac{4}{10} = \frac{2}{5}, \qquad p_X(1) = \frac{6}{10} = \frac{3}{5}.$$

The marginal PMF of Y is

$$p_Y(0) = \frac{2}{10} = \frac{1}{5}, \qquad p_Y(1) = \frac{4}{10} = \frac{2}{5}, \qquad p_Y(2) = \frac{4}{10} = \frac{2}{5}.$$

Each marginal is non-negative and sums to one. Independence would require $p_{X,Y}(x,y) = p_X(x)p_Y(y)$ for every point in the support. At $(x,y) = (1,2)$,

$$p_{X,Y}(1,2) = \frac{3}{10}, \qquad p_X(1)p_Y(2) = \frac{3}{5}\frac{2}{5} = \frac{6}{25}.$$

These values differ, so X and Y are not independent.

Final result. The marginal PMFs are shown above, and the variables are not independent.

Marks (6). Correct p_X, 1; correct p_Y, 2; validity checks, 1; valid factorisation comparison, 1; independence conclusion, 1.

Common mistakes. Summing along the wrong dimension; omitting support values with positive probability; declaring independence after checking only that both marginals sum to one.

`W05-Q06-PB (S5)`

Method and conditions. Every $y \in \{0, 1, 2\}$ has positive marginal probability, so divide its joint-PMF column by $p_Y(y)$.

The conditional PMFs are

	$p_{X\mid Y}(0 \mid y)$	$p_{X\mid Y}(1 \mid y)$
$y = 0$	$1/2$	$1/2$
$y = 1$	$1/2$	$1/2$
$y = 2$	$1/4$	$3/4$

on the conditional support $x \in \{0, 1\}$. Because X is Bernoulli conditional on each y,

$$\mathbb{E}[X \mid Y] = \begin{cases} 1/2, & Y = 0, \\ 1/2, & Y = 1, \\ 3/4, & Y = 2. \end{cases}$$

The law of iterated expectation gives

$$\begin{aligned} \mathbb{E}[\mathbb{E}[X \mid Y]] &= \frac{1}{2}\frac{1}{5} + \frac{1}{2}\frac{2}{5} + \frac{3}{4}\frac{2}{5} \\ &= \frac{1}{10} + \frac{2}{10} + \frac{3}{10} \\ &= \frac{3}{5}. \end{aligned}$$

From the marginal PMF, $\mathbb{E}[X] = \mathbb{P}(X = 1) = 3/5$, so the two calculations agree.

Final result. The conditional PMFs and conditional expectation are shown above, and $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X] = 3/5$.

Marks (7). Conditional PMFs for $y = 0, 1, 2$; conditional PMF for $y = 2, 1$; conditional support/positive denominators, 1; three conditional means, 1; iterated-expectation calculation, 1; marginal reconciliation, 1.

Common mistakes. Dividing by $p_X(x)$ rather than $p_Y(y)$; reporting one conditional mean instead of the random variable $\mathbb{E}[X \mid Y]$; weighting conditional means equally rather than by $p_Y(y)$.

`W05-Q07-PA (S5)`

Method and conditions. Normalise and integrate over the triangular support $0 < x < y < 1$.

$$1 = \int_0^1 \int_0^y c \, \mathrm{d}x \, \mathrm{d}y = \frac{c}{2},$$

so $c = 2$.

For $0 < x < 1$, the admissible values of y are $x < y < 1$, hence

$$f_X(x) = \int_x^1 2 \, \mathrm{d}y = 2(1 - x).$$

For $0 < y < 1$, the admissible values of x are $0 < x < y$, hence

$$f_Y(y) = \int_0^y 2\,\mathrm{d}x = 2y.$$

Both densities are zero outside $(0,1)$ and each integrates to one. Because $f_Y(y) = 2y > 0$ for $0 < y < 1$,

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{1}{y}, \qquad 0 < x < y,$$

for $0 < y < 1$, and it is zero outside the conditional support.

Final result. $c = 2$, $f_X(x) = 2(1-x)$ on $0 < x < 1$, $f_Y(y) = 2y$ on $0 < y < 1$, and $f_{X|Y}(x \mid y) = 1/y$ on $0 < x < y$ for $0 < y < 1$.

Marks (5). Normalisation and c, 1; correct f_X and support, 1; correct f_Y and support, 1; conditional density, 1; conditional support and denominator condition, 1.

Common mistakes. Integrating over a rectangle instead of the triangle; reversing the limits for a marginal; omitting the condition $f_Y(y) > 0$ or the conditional support.

`W05-Q07-PB (S5)`

Method and conditions. Use the conditional density on $0 < x < y$, then apply iterated expectation with the marginal density of Y.

$$\mathbb{E}[X \mid Y = y] = \int_0^y x\frac{1}{y}\,\mathrm{d}x = \frac{y}{2}, \qquad 0 < y < 1.$$

Also,

$$\mathbb{E}[Y] = \int_0^1 y(2y)\,\mathrm{d}y = \frac{2}{3}.$$

Therefore,

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]] = \frac{1}{2}\mathbb{E}[Y] = \frac{1}{3}.$$

Directly from the joint density,

$$\mathbb{E}[X] = \int_0^1 \int_0^y 2x\,\mathrm{d}x\,\mathrm{d}y = \int_0^1 y^2\,\mathrm{d}y = \frac{1}{3}.$$

Final result. $\mathbb{E}[X \mid Y = y] = y/2$ for $0 < y < 1$, and $\mathbb{E}[X] = 1/3$ by both methods.

Marks (4). Conditional expectation and support, 1; marginal mean of Y, 1; iterated-expectation result, 1; direct verification, 1.

Common mistakes. Integrating the conditional density over $0 < x < 1$; using an unweighted average of conditional means; changing the triangular limits in the direct check.

`W05-Q08-PA (S5)`

Method and conditions. Conditional on either value of Y, X is Bernoulli. Apply

$$\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(\mathbb{E}[X \mid Y]).$$

The conditional means are

$$\mathbb{E}[X \mid Y=0]=\frac{1}{4}, \qquad \mathbb{E}[X \mid Y=1]=\frac{3}{4},$$

and both conditional variances are

$$\frac{1}{4}\frac{3}{4}=\frac{3}{16}.$$

Therefore,

$$\mathbb{E}[\mathrm{Var}(X \mid Y)]=\frac{3}{16}.$$

As a random variable,

$$\mathbb{E}[X \mid Y]=\frac{1}{4}+\frac{1}{2}Y.$$

Since $\mathrm{Var}(Y)=(1/3)(2/3)=2/9$,

$$\mathrm{Var}(\mathbb{E}[X \mid Y])=\frac{1}{4}\,\mathrm{Var}(Y)=\frac{1}{18}.$$

It follows that

$$\mathrm{Var}(X)=\frac{3}{16}+\frac{1}{18}=\frac{35}{144}.$$

For verification, total probability gives

$$\mathbb{P}(X=1)=\frac{2}{3}\frac{1}{4}+\frac{1}{3}\frac{3}{4}=\frac{5}{12}.$$

Thus $X \sim \mathrm{Bernoulli}(5/12)$ marginally, and

$$\mathrm{Var}(X)=\frac{5}{12}\frac{7}{12}=\frac{35}{144}.$$

Final result. The within-group component is $3/16$, the between-conditional-means component is $1/18$, and $\mathrm{Var}(X)=35/144$, confirmed from the marginal Bernoulli distribution.

Marks (10). Conditional means and variances, 2; expected conditional variance, 1; conditional-mean random variable, 1; variance of conditional means, 2; total-variance sum, 1; marginal success probability, 2; marginal variance verification, 1.

Common mistakes. Using only the expected conditional variance; computing $\mathrm{Var}(\mathbb{E}[X \mid Y])$ without squaring the coefficient; averaging conditional success probabilities without the probabilities of Y; adding the two checks as if they were separate variances.

`W05-Q09-PA` (M5 retrieval)

Method and conditions. Solve the characteristic equation, then find each eigenspace. A real 2×2 matrix with two linearly independent eigenvectors is diagonalizable.

$$\det(\mathbf{K} - \lambda \mathbf{I}_2) = (2-\lambda)^2 - 1 = (\lambda - 3)(\lambda - 1).$$

For $\lambda = 3$, an eigenvector basis is

$$\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}.$$

For $\lambda = 1$, an eigenvector basis is

$$\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

The two eigenvectors are linearly independent.

One valid diagonalisation is

$$\mathbf{K} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}, \qquad \mathbf{P} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad \mathbf{D} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

Final result. The eigenvalues are 3 and 1, with the eigenspace bases above; $\mathbf{K}$ is diagonalizable as displayed.

Marks (4). Eigenvalues, 1; two eigenspace bases, 1; diagonalizability conclusion, 1; valid $\mathbf{P}$ and matching $\mathbf{D}$, 1.

Common mistakes. Solving the entrywise equation instead of the determinant; giving the zero vector as an eigenvector; claiming distinct eigenvalues without identifying eigenvectors; mismatching eigenvector columns and diagonal entries.

`W05-Q10-PA` **(S2 retrieval)**

Method and conditions. The events A, A^c form a partition. Use total probability, then Bayes' rule; the resulting $\mathbb{P}(C)$ is positive.

$$\begin{aligned} \mathbb{P}(C) &= \mathbb{P}(C \mid A)\mathbb{P}(A) + \mathbb{P}(C \mid A^c)\mathbb{P}(A^c) \\ &= \frac{3}{4}\frac{1}{4} + \frac{1}{4}\frac{3}{4} = \frac{3}{8}. \end{aligned}$$

Bayes' rule gives

$$\mathbb{P}(A \mid C) = \frac{\mathbb{P}(C \mid A)\mathbb{P}(A)}{\mathbb{P}(C)} = \frac{3/16}{3/8} = \frac{1}{2}.$$

Final result. $\mathbb{P}(C) = 3/8$ and $\mathbb{P}(A \mid C) = 1/2$.

Marks (4). Complement probability, 1; total-probability setup and result, 1; Bayes setup, 1; posterior, 1.

Common mistakes. Omitting the A^c branch; reversing the requested conditional without Bayes' rule; dividing by $\mathbb{P}(A)$ rather than $\mathbb{P}(C)$.

`W05-Q10-PB` **(S4 retrieval)**

Method and conditions. Use linearity of expectation and the variance rule $\operatorname{Var}(a + bU) = b^2 \operatorname{Var}(U)$.

$$\mathbb{E}[V] = 4 - 2\mathbb{E}[U] = 4 - 4 = 0,$$

whereas

$$\operatorname{Var}(V) = (-2)^2 \operatorname{Var}(U) = 4(3) = 12.$$

Final result and interpretation. $\mathbb{E}[V] = 0$ and $\operatorname{Var}(V) = 12$. Adding a constant shifts every value and hence the mean, but it does not change deviations from the mean. Scaling by -2 multiplies those deviations by -2, so variance is multiplied by four.

Marks (3). Correct mean, 1; correct variance, 1; explanation, 1.

Common mistakes. Adding four to the variance; multiplying variance by -2 rather than by four; claiming that a negative scale gives negative variance.

Self-test solutions and error check

Solutions and error check

Solution to `W05-ST01-PA (M7)` Method and conditions. The function is continuously differentiable. Check that the partial derivative with respect to the endogenous variable is non-zero.

$$F_x(x, \theta) = e^x(1 + x), \qquad F_\theta(x, \theta) = 1.$$

At $(0, 1)$, $F_x = 1 \neq 0$. Therefore,

$$\left.\frac{\mathrm{d}x}{\mathrm{d}\theta}\right|_{(0,1)} = -\frac{F_\theta}{F_x} = -1.$$

Final result. The local condition holds and $\mathrm{d}x/\mathrm{d}\theta = -1$ at the point.

Marks (5). Differentiability and correct local condition, 1; correct F_x, 1; verified non-zero value, 1; correct implicit formula, 1; final derivative, 1.

Common mistakes. Differentiating xe^x without the product rule; checking $F_\theta \neq 0$ instead of $F_x \neq 0$; omitting the minus sign.

Solution to `W05-ST02-PA (M8)` Method and conditions. The matrix is symmetric, so use its eigenvalue signs. The Hessian of the quadratic function is $2\mathbf{A}$.

$$\det(\mathbf{A} - \lambda \mathbf{I}_2) = (1 - \lambda)^2 - 4 = (\lambda - 3)(\lambda + 1).$$

The eigenvalues are 3 and -1. Since they have opposite signs, the quadratic form is indefinite. The Hessian $2\mathbf{A}$ also has one positive and one negative eigenvalue, so it is neither positive semidefinite nor negative semidefinite.

Final result. The form is indefinite, and the function q is neither convex nor concave on $\mathbb{R}^2$.

Marks (5). Characteristic equation or equivalent test, 1; eigenvalues, 1; indefinite classification, 1; Hessian relation, 1; neither-convex-nor- concave conclusion, 1.

Common mistakes. Calling a positive trace positive definiteness; checking only the determinant without interpreting its sign; forgetting the factor two in the Hessian while reaching an inconsistent curvature conclusion.

Solution to `W05-ST03-PA (S5)` Method and conditions. Both values of Y have positive marginal probability. Divide each column by its marginal, then average the conditional means using the marginal PMF of Y.

$$\mathbb{P}(Y = 0) = \mathbb{P}(Y = 1) = \frac{1}{2}.$$

Thus

$$\mathbb{P}(X=1 \mid Y=0)=\frac{1/6}{1/2}=\frac{1}{3}, \qquad \mathbb{P}(X=1 \mid Y=1)=\frac{1/3}{1/2}=\frac{2}{3}.$$

Since X takes values zero and one,

$$\mathbb{E}[X \mid Y]=\begin{cases} 1/3, & Y=0, \\ 2/3, & Y=1. \end{cases}$$

Consequently,

$$\mathbb{E}[\mathbb{E}[X \mid Y]]=\frac{1}{3}\frac{1}{2}+\frac{2}{3}\frac{1}{2}=\frac{1}{2}.$$

The marginal probability $\mathbb{P}(X=1)=1/6+1/3=1/2$, so $\mathbb{E}[X]=1/2$ and the identity is verified.

Final result. The conditional expectation takes values $1/3$ and $2/3$, and its expectation equals $\mathbb{E}[X]=1/2$.

Marks (5). Marginal PMF of Y, 1; two conditional probabilities, 1; conditional-expectation random variable, 1; iterated calculation, 1; marginal check, 1.

Common mistakes. Dividing by a row marginal; reporting only one conditional value; weighting the conditional means without $\mathbb{P}(Y=y)$.

Solution to `W05-ST04-PA (S5)` Method and conditions. Apply the law of iterated expectation and then separate the within- and between-group terms in total variance.

$$\mathbb{E}[X]=\mathbb{E}[\mathbb{E}[X \mid Y]]=\mathbb{E}[1+Y]=1+\frac{1}{2}=\frac{3}{2}.$$

The within-group component is

$$\mathbb{E}[\mathrm{Var}(X \mid Y)]=\mathbb{E}[2]=2.$$

The between-group component is

$$\mathrm{Var}(\mathbb{E}[X \mid Y])=\mathrm{Var}(1+Y)=\mathrm{Var}(Y)=\frac{1}{2}\frac{1}{2}=\frac{1}{4}.$$

Therefore,

$$\mathrm{Var}(X)=2+\frac{1}{4}=\frac{9}{4}.$$

Final result. $\mathbb{E}[X]=3/2$; the two variance components are 2 and $1/4$, giving $\mathrm{Var}(X)=9/4$.

Marks (5). Iterated-expectation setup, 1; mean, 1; expected conditional variance, 1; variance of the conditional mean, 1; total variance, 1.

Common mistakes. Adding conditional variances without expectation; omitting the between-group component; treating $\mathrm{Var}(1+Y)$ as $1+\mathrm{Var}(Y)$.

Record the first cause of each error under `M7`, `M8` or `S5`, then reattempt that part from a blank page in two or three days.